



Analyze Quiz

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Responses

Date Submitted

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Score

30/40

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75%

1.

All students at a certain high school participate in at least one of three different sports: cricket, soccer, and volleyball. The following is known about their participation:

- None of the students that play cricket also play soccer.
- Some of the students that play soccer also play volleyball.
- Every student that plays volleyball plays at least one other sport.

Thomas is a student at this high school, and we know that he plays soccer.

1. Which one of the following statements is true?



We know that he plays volleyball but not cricket



We know that he only plays one sport



We know that he might also play cricket



We know that he might also play volleyball



0/1

Question Feedback:

If Thomas plays soccer, then we know that he might play volleyball as per the second statement. He cannot play cricket as that would contradict the first statement. It cannot be option (a) as it is not certain that he plays volleyball.

Thus, the answer is (d).

2.

Three students –Tom, Richard, Tand Harry, are suspects in an investigation into who vandalised the school walls by drawing graffiti. Only one of them is guilty. When questioned they gave the following statements:

Tom: Harry drew the graffiti.

Richard: I drew the graffiti

Harry: Richard drew the graffiti.

Only one of the three is telling the truth and the other two are lying.

1/1

1. Which of the following is false?



Richard did not draw the graffiti



Harry is lying



Harry drew the graffiti



Tom is lying



Question Feedback:

When considering the statements, you must think on who are the two lying. If we take that Tom is lying, then it becomes a paradoxical scenario. Richard and Harry are both saying that Richard drew the graffiti, however, only one person can be telling the truth. Hence, Tom is telling the truth and Richard and Harry are lying.

This means that Tom cannot be lying and so it is answer (d).

3.

If Daniel comes first in a subject this year, his dad will give him \$400. If Daniel doesn't study 5 hours a day, he can't come first in a subject.

If Daniel has \$400, he can buy the new video game. If Daniel doesn't buy the video game this year, he will have to wait 5 years to get the next version of the video game. Daniel will only receive \$400 if he comes first in a subject and his dad gives it to him.

1/1

1. Which one of the following outcomes is not possible?



Daniel studied 5 hours a day and didn't come first in a subject



Daniel studied 5 hours a day and came first in a subject



Daniel bought the video game without coming first in a subject



Daniel studied 10 hours a day and didn't come first in a subject

Question Feedback:

Out of the given options, answer (c) is impossible. This is because Daniel will not be able to get the \$400 if he doesn't come first in a subject and thus will not be able to buy the video game.

This is answer (c).

4.

To be eligible for a promotion at Sanscorp an employee must have:

- A 95% attendance rate
- Worked in the company for 10 years
- Led at least 3 successful projects
- Worked under a senior manager for at least 3 years
- Completed a course on conflict resolution outside work hours

Alaina: I've worked at Sanscorp for 12 years, my last two projects have been my only successful projects and I have worked under senior manager Darius for the last 3 projects. I am eligible for a promotion.

1/1

1. Which of the following do we know for certain is a reason why Alaina is not eligible for a promotion?



Alaina's attendance rate is less than 95%



Alaina's has not led at least 3 successful projects



Alaina has not led any projects



Alaina worked under a senior manager for less than 3 years

Question Feedback:

From Alaina's statement, she states that she has only led 2 successful projects. Thus, we know for certain that she has not led 3 successful projects, which disqualifies her from getting the promotion.

This is answer (b).

5.

At a certain school, students are allowed to participate in either fencing or pottery depending on their performance in other areas of school.

The following table shows the conditions required to participate in either fencing or pottery.

Fencing	> 90% attendance
	> 80% in PE
	Failed < 1 subject
Pottery	> 80% attendance
	> 80% in Visual Arts
	Failed < 2 subjects

The following table shows the performance of three students that attend the school.

	Adam	Barry	Cathy
Attendance (%)	95	82	96
PE (%)	54	72	99
Visual Arts (%)	93	78	91
Failed Subjects	0	2	1

1/1

1. Who can participate in Pottery?

☐

Adam

☐

Barry

☒

Cathy and Adam

✓

☐

No one

Question Feedback:

Barry is not qualified as he does not have over 80% in Visual Arts.

Cathy and Adam however, both meet the requirements needed to participate in pottery.

This is answer (c).

6.

"In a certain primary school football league, if a team loses every game in one year, it is moved to a lower division the next year where the team can then play weaker opponents. Richmond Public School's football team failed to win a single game last year and so the team fired the head coach. Ever since the team hired a new coach, the team has not lost a single game."

Brian, a student at Richmond Public School: Our football team has not lost a single game since we changed our head coach. Our new coach must be the best coach in the whole tournament.

Mike, a student from a neighbouring school: The only reason you guys are even winning is because your competition is worse this year. Your new coach isn't that good.

0/1

1. If the quoted passage is true, whose reasoning is correct?

☐

Brian only

☒

Mike only

✗

☐

Both Brian and Mike

☐

Neither Brian nor Mike

✓

Question Feedback:

Brian's statement is incorrect as the team got put into a lower division and are versing people who are weaker and so the team is more likely to win more games now.

Mike's statement is incorrect as it could also be the guidance of the new coach who directs his players well which is contributing to the team's success.

Since both statements are incorrect, the answer is (d).

7.

A certain farm has snakes, horses, and chickens. Snakes have no legs, horses have 4 legs and chickens have 2 legs. There are 10 animals altogether and 10 legs in total.

1. What is the highest possible number of horses that could be at the farm?

1/1

0

1

2



3

Question Feedback:

We know that the farm has at least 1 of each animal. If we want to maximise the number of horses, then we would try to maximise the legs which belonged to the horses.

There are 10 legs, which means that the maximum number of horses there could be is 2 horses. This would also mean there is only 1 chicken, and the other 7 animals are snakes.

This is answer (c).

8.

4 athletes —David, Eliot, James, and Samuel— ran a 100m sprint together.

Only one person finished after David.

One person finished ahead of Eliot.

James finished after David.

1. Where did Samuel finish?

1/1

First



Second

Third

Fourth

Question Feedback:

Order: FIRST – Samuel – Eliot – David – James – LAST

This shows that Samuel came first which is answer (a).

9.

Sally's father says that children should not lift weights as it will cause them to be short when they are adults. As a result, Sally's father has banned her from lifting weights until she has grown to her full adult height.

1. Which of the following statements, if true, most weakens Sally's father's argument?

☐

Children's growth plates close once they become adults

☒

On average, those who weightlifted regularly as children are the same height as the general population

✓

☐

Most professional and elite weightlifters that Sally knows are short

☐

Weightlifting releases a hormone which improves athletic ability

Question Feedback:

The argument is that weightlifting as a child is detrimental to height growth. Option (a) is irrelevant as we are not given any connection between growth plates and height. Option (c) is also irrelevant. Option (d) is irrelevant to the argument. as it relates to growth in athletic ability rather than height. Option (b) weakens the argument by showing a trend that weightlifting doesn't affect height growth.

Hence, answer (b) best helps in weakening Sally's father's argument.

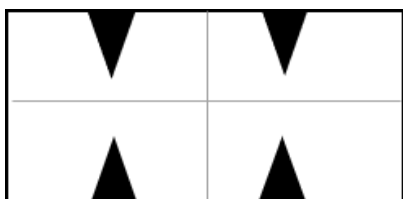
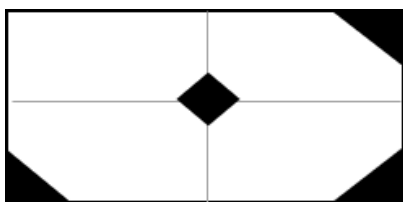
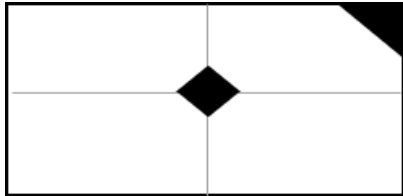
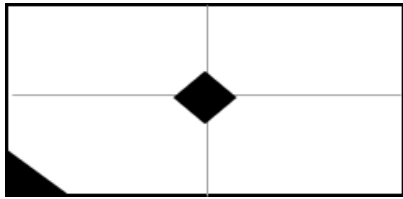
10.

A rectangular piece of paper is folded exactly in half and then folded in half again. A single cut-out of the folded paper is taken. The piece of paper is then unfolded twice.

1. Which of the following could be the piece of paper when unfolded twice?

1/1

1/1

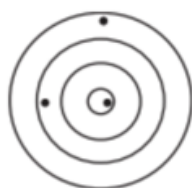
**Question Feedback:**

Since it is folded twice and a single cut is done then when it is opened up again, the hole of the cut should be visible in all 4 quadrants of the opened paper. Other options had multiple holes in a single quadrant which is not possible with just one cut.

The only option which did this is answer (d), and so is the only correct answer.

11.

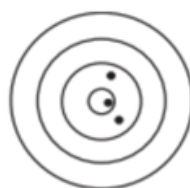
Andrew and his friends competed in a darts contest. Each person had three shots at the target:



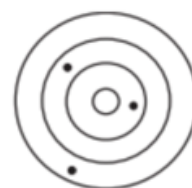
Andrew



Chris



Richie



Kyle

- Andrew scored 14 points.

- Chris missed one shot and scored 6 points.
- Richie scored 22 points.

All four zones of the target are worth a different whole number of points. The bullseye is worth the most points, and each zone is worth fewer points than the one inside it.

1. What did Kyle score?

☐


☒


☐

☐

Question Feedback:

Method 1:

Andrew = 1x bullseye, 1x third ring, 1x fourth ring

Kyle = 1x second ring, 1x third ring, 1x fourth ring

Their third ring and fourth ring scores are the same, so the only difference is that Andrew got a bullseye, while Kyle got the second ring, which is worth less. This means that Kyle must have scored below Andrew's points, and thus option (a) is the only answer that represents this.

Method 2:

Since Chris scored 2 in the same zone you can find out how many points the 2nd circle is worth.

Chris's score / 2 = Circle 2 points

$6/2 = 3$.

Andrew scored in the middle circle, the 2nd circle, and the outside circle. We know that the 2nd circle is worth 3 points therefore the total of Andrew's other 2 shots is 11. Since we know that the outside circle is worth less points than the 2nd circle it must be worth either 1 or 2 points. This also means that the centre most circle must be worth either 9 or 10 points.

Richie has scored 22 points and hit 2 in the 3rd circle and 1 in the middle circle. From above we know that the middle circle is worth either 9 or 10 points.

If the middle circle is 9 points then the 3rd circle must be worth 5.5 points.

$22 - 9 = 13$

$13/2 = 6.5$

0/1

This is not possible as each circle is worth a different whole number of points. Therefore the centre circle is worth 10 points, which means that the 3rd circle is worth 6 points.

Overall, the circles from outside to in are worth 1,3,6 and 10 points respectively.

Kyle hit 1 arrow in each circle except the middle circle.

His total is $1 + 3 + 6 = 10$.

This is answer (a).

12.

Seven students participate in a shotput competition where they push (rather than throw) a ball as far as they can.

The table below shows the distances each student pushed in each of their attempts.

<i>Student Number</i>	<i>Attempt 1 (m)</i>	<i>Attempt 2 (m)</i>
1	7	9
2	5	10.2
3	6	8.5
4	9	8
5	11	9.5
6	5	8
7	9.5	6

The final score of each student is the average of the distances for both their attempts.

1. Which student came 4th in the competition?

☐

Student Number 2

☐

Student Number 3

☒

Student Number 6

✗

☐

Student Number 7

✓

Question Feedback:

Find the total distance for each student:

Student 1: $7 + 9 = 16$ m

Student 2: $5 + 10.2 = 15.2$ m

0/1

Student 3: $6 + 8.5 = 14.5$ m

Student 4: $9 + 8 = 17$ m

Student 5: $11 + 9.5 = 20.5$ m

Student 6: $5 + 8 = 13$ m

Student 7: $9.5 + 6 = 15.5$ m

We then need to rank the students by their total distance to find out who's average distance is 4th place (average distance is their total distance divided by 2 but calculating this takes extra time that is not needed!):

1. Student 5: 20.5 m

2. Student 4: 17 m

3. Student 1: 16 m

4. Student 7: 15.5 m

5. Student 2: 15.2 m

6. Student 3: 14.5 m

7. Student 6: 13 m

From this we can see that the student who came 4th is Student Number 7 with a total distance of 15.5m and average of 7.75m which is answer (d).

13.

Recruiters for a talent agency attend a school talent show to select a candidate to train to be a performer. They assess contestants based on their dancing and vocal abilities. The contestants are Albert, Suzy, Sam, and Brandon.

Albert's vocals are better than Brandon's, but Brandon's vocals are better than Suzy's. Suzy's dancing isn't as good as Albert's.

Brandon's dancing is better than Albert's, but it is not as good as Sam's dancing. Albert does not have the best vocals.

1. If the recruiters end up selecting the contestant with the best vocals, which of the four contestants did the recruiters choose?

☐

Albert

☐

Suzy

1/1

**Question Feedback:**

Order Singing: BEST – Sam – Albert – Brandon – Suzy - WORST

Order Dancing: BEST – Sam – Brandon – Albert – Suzy – WORST

This shows that the best vocals belong to Sam which is answer (c).

14.

Four friends Sally, Fred, David, and Adam ran in their school marathon.

- Sally finished ahead of Fred.
- David finished behind Sally.
- Adam did not finish last.

1. Which one of the following MUST be true?

**Question Feedback:**

Option (a) may not be true as Adam not finishing last could mean that he beat Sally and came first.

Option (b) may not be true as David could be ahead of Fred when they finished and one of them could be last.

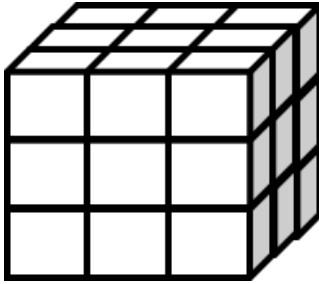
Option (c) may not be true as Adam not being last would mean that he may have only beaten one of them and so he could be ahead of Fred but behind David which makes this statement false.

This shows that only answer (d) is the correct answer.

1/1

15.

Jeremy has a large cube made of many small, white cubes as shown below.



He dips the large cube into a bucket of red paint. He then removes all small cubes that have red paint on their faces. He dips the remaining small cubes into a bucket of blue paint.

1/1

1. How many of the small cubes have blue paint on their faces?

☐

0

☒

1

☐

2

☐

3

Question Feedback:

This cube is a $3 \times 3 \times 3$. This means that it is made up of 27 small cubes. When the large cube is dipped into the red paint then all exterior cubes must be removed. This is 26 of the small cubes.

This means that the only cube with blue paint is the 1 cube in the very centre.

This is answer (b).

16.

At an animal enclosure there are kangaroos, tigers, and owls. There is at least one of each animal in the enclosure. Kangaroos have two legs, tigers have four legs and owls have two legs. In total there are 12 animals and 28 legs.

1. What is the maximum numbers of owls that could be in the enclosure?

0/1

7

8

9



10

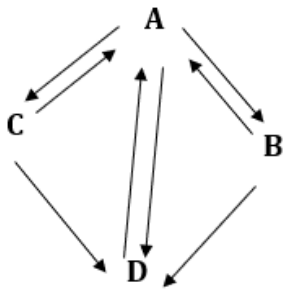
**Question Feedback:**

We need at least one of each animal, as so we must have at least one kangaroo and one tiger. This makes it so we have 10 owls which comes to a total of 12 animals and 26 legs. To reach 28 legs, we need to increase the number of tigers. Thus with 2 tigers, 1 kangaroo and 9 owls we can get 12 animals and 28 legs.

This is answer (c).

17.

Marco is walking along the paths on his school campus. Marco starts at point A and wants to reach point D. Before reaching D, he must visit B or C at least once. Marco must follow the arrows in the direction shown. He may visit the same point more than once, but he may use each arrow only once. Once he reaches D, his journey ends. All possible paths connecting points A, B, C and D on his school campus are shown in the diagram below:



1. How many different ways can Marco travel from point A to point D?

0/1

4



5

☐
6☐
8**Question Feedback:**

From the original question we know that Marco must pass points B or C before reaching point D. This means that he cannot travel directly from A to D at the beginning.

Paths:

A → B → D

A → C → D

A → B → A → D

A → C → A → D

A → B → A → C → D

A → C → A → B → D

A → B → A → C → A → D

A → C → A → B → A → D

This shows that there are 8 paths which is answer (d).

18.

A school employs a security guard to prevent crime from occurring at school during the day. If a person is found committing a crime at school, they are detained and then expelled. The security guard is paid \$50,000 per year. There has not been a crime committed at the school in the past 3 years.

Sean: "The school is just wasting money on a security guard, no one at the school is committing crimes so he isn't doing anything useful."

1. Which of the following shows the mistake Sean has made?

☐
The school may have excess funds which they must spend on a security guard☐
The security guard may not be doing their job properly, so the school does not know of the crimes being committed

1/1



The security guard's presence may be the reason that no crimes are being committed



The school may not be able to call the police in time when a crime occurs so having a security guard is necessary for emergencies

Question Feedback:

The mistake in Sean's reasoning is that he has wrongly assumed that the security guard isn't doing anything. It could possibly be that his presence has been the driving force in stopping the crime in the school.

This is answer (c).

19.

"A survey was conducted by interviewing successful business owners and asking them what kind of education they had. 200 successful business owners were interviewed in total and only 20 of them did go to university. However, all of them graduated high school."

Tobi: Business owners are more likely to be successful if they do not go to university.

Tyrone: Most successful business owners in this study skipped university and only focused on their businesses since they graduated high school.

1. If the quoted passage is true, whose reasoning is correct?



Tobi only



Tyrone only



Both Tobi and Tyrone



Neither Tobi nor Tyrone

**Question Feedback:**

Tobi's statement is incorrect as not going to university won't guarantee that a person is more likely to become a successful business owner and it could be that they are also more likely to fail. The passage does not provide any information on this matter so we cannot make this assumption.

Tyrone's statement is incorrect as we do not know if they focused on their business after graduating high school. They could have also focused on other things apart from their business.

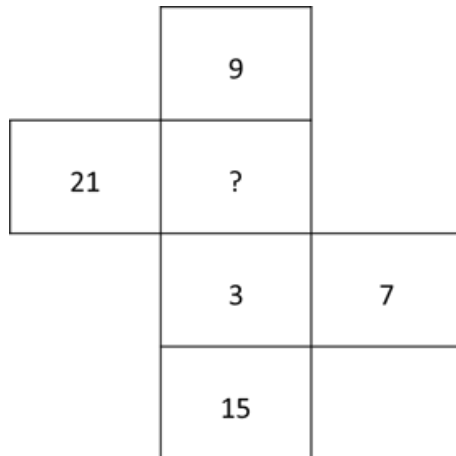
0/1

Since both statements are incorrect, the answer is (d).

20.

Jamie's Cube Store makes cubes where all pairs of opposite faces on the cube have a number following a certain rule.

A net of a cube from Jamie's Cube Store is shown below. One face's number is missing.



1. Which of the following numbers could replace the question mark?

☐

3

☐

17

☒

5

✓

☐

20

Question Feedback:

Look at the opposing faces of sides which we know the number of already. We can see that the 3 is opposite the 9 and the 7 is opposite the 21. This rule is that one side is 3 times the other side.

Since the last side is 15, we should be looking for either 5 or 45. Only 5 is possible from the answer options.

This is answer (c).

21.

"Derek plays games on his laptop during the summer. If the temperature outside is too high, his laptop will stop working and he will have to buy a new laptop. If Derek's laptop is broken, he will have to work extra hours at his job to buy a new laptop. Claudia and Brian notice that Derek has been working extra hours at his job this summer."

Claudia: Derek has been working extra hours at his job so he must be playing video games.

Brian: If Derek is playing games on his laptop this summer, he may have to buy a new laptop.

1/1

1. If the quoted passage is true, whose reasoning is correct?

☐

Claudia only

☒

Brian only

☐

Both Claudia and Brian

☐

Neither Claudia nor Brian

Question Feedback:

Claudia's statement is incorrect as the temperature outside could have been hot enough to damage his laptop which means that he needs to work extra hours in order to buy a new laptop.

Brian's statement is correct, as the temperature increase could break his laptop and result in him needing to buy a new laptop.

Since only Brian is correct, the answer is (b).

22.

During the summer, Pablo opened an ice cream truck near the beach. The temperature has been rising and so has Pablo's ice cream sales. Pablo has noticed that ever since he opened his ice cream truck there have been more and more shark attacks at the nearby beach. Pablo does not wish for any more people to be harmed so he closes his ice cream truck and searches for something else to sell.

1. Which of the following shows the mistake Pablo has made?

1/1

Even if he closes his ice cream truck people may still be attacked by other sea creatures such as sting rays

If Pablo sells his ice cream at higher prices, less people may buy it and he may not have to close the truck

The shark attacks may have increased because it was summer and there were more people at the beach, not because of Pablo's ice cream truck



The shark attacks may increase even more if he begins selling something else

Question Feedback:

The mistake in Pablo's reasoning is that he has wrongly assumed that his ice cream sales caused more people to come to the beach. Instead, it could just be summer which increases the number of sharks coming into feed.

This is answer (c).

23.

"At a certain school, teachers who wish to be a part of the Parent-Teacher Association must apply for the position by March 15th. Last year, all 25 teachers at the school applied to be a part of the Parent-Teacher Association. This year only 20 teachers applied."

Lauren: Five teachers must have missed the deadline to apply this year.

Lew: All 20 teachers that applied this year also applied last year.

1. If the quoted passage is true, whose reasoning is correct?

Lauren only

Lew only

Both Lauren and Lew

Neither Lauren nor Lew



1/1

Question Feedback:

Lauren's statement is incorrect as the other 5 teachers may not have wanted to go back to the association this year and opt to not enter in the first place.

Lew's statement is incorrect as there may be new teachers that have been hired at the school.

Since both statements are incorrect, the answer is (d).

24.

Peter has created a mathematical code where he uses various symbols in place of the four basic mathematical operators (+, -, ×, ÷). The symbols satisfy the following set of equations:

- $5\%3 = 2$
- $4^3 = 12$
- $2^2 = 4$
- $9\%6 = 3$
- $10\%4 = 6$

1. Calculate the following: $(7^8)\%9$



47



36



25



14

Question Feedback:

From the equations we can figure out what the symbols mean:

$$5 - 3 = 2$$

$$\% = -$$

$$4 \times 3 = 12$$

$$^ = \times$$

$$(7^8)\%9 = (7 \times 8) - 9$$

$$= 56 - 9$$

$$= 47$$

1/1

This is answer (a).

25.

An ice hockey recruiter attends hockey league matches for 12-year-olds to determine what affects hockey players performance. He picks three teams randomly from the league. He gives the first team a glass of milk to drink before they play, he gives another team a glass of water to drink before they play, and he gives the final team a glass of soda to drink before they play. The team that drank soda played the best, while the team that drank milk played the worst.

Recruiter: "This experiment indicates that that drinking a glass of soda before a game makes players play better."

1. Which of the following shows the mistake the recruiter has made?



The team who drank milk might drink more soda at home



The team who drank water would play better if they drank more water



The team who drank milk may play better at turf hockey



The team that played better may have been more skilled prior to the experiment



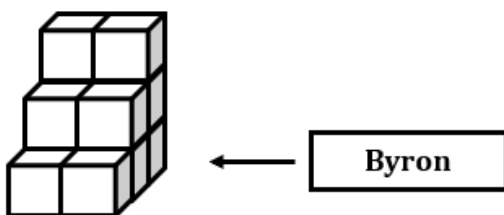
Question Feedback:

The mistake in the recruiter's statement is that he has wrongly assumed that what the players are drinking before a game determines how they will play during the match. However, the team could have been better at the game and trained harder to get better than the other teams.

This is answer (d).

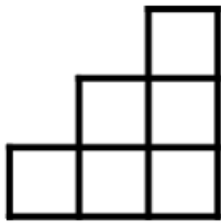
26.

Byron is facing the right of a large stack of cubes. This is shown in the diagram below:



Currently his view looks like this:

1/1



1. What is the minimum number of cubes Bryon needs to remove from the stack of cubes to change his view?

☐

1

☒

2


☐

3

☐

4

Question Feedback:

Since the blocks are all 2 blocks in depth from his view, as soon as Byron moves the two blocks from the top or bottom then he will have a completely new shape with a different front view.

This is answer (b).

27.

"At a certain paper company, management is considering individuals for a promotion. An individual must have worked at their position for 3 years and led a successful project to get a promotion. If an individual is promoted, they cannot be promoted the following year."

Penny: In my section, there are 20 employees that have worked at their position for 3 years and 10 employees that have led a successful project. Therefore, 30 people must be getting promoted.

Albert: In my section, there are 10 employees that have worked at their position for 3 years and 7 employees that have led a successful project. Therefore, 7 people are getting promoted because only 7 people could have both worked in their position for 3 years and led a successful project this year.

1. If the quoted passage is true, whose reasoning is correct?

1/1



Penny only



Albert only



Both Penny and Albert



Neither Penny nor Albert



Question Feedback:

Penny's statement is incorrect as the requirement is that the person must have worked for 3 years and led a successful project. Only having one is not enough for promotion. Hence, only a maximum of 10 employees would be available for promotion.

Albert's statement is incorrect as satisfying both criteria only puts the person up as eligible for the promotion, but that does not mean that every single one of them will get the promotion.

Since both statements are incorrect, the answer is (d).

28.

A teacher wants to know if students learn mathematics better in the morning or in the evening. The teacher has two mathematics classes –Class A in the morning and Class B in the evening. The teacher gives both classes a test on mathematics and calculates the average score for each class. The average score for Class A is higher than the average score for Class B. The teacher concludes that students learn mathematics better in the morning.

1. If the above information is true, which of the following shows the mistake the teacher has made?



Students in Classes A and B may be smarter than the average student



Students in Classes A and B may have a higher workload than the average student



Students in Class A may be smarter on average than students in class B



Students in Classes A and B both hate mathematics so it doesn't matter if they learn in the morning or in the evening

Question Feedback:

1/1

The mistake in the teacher's reasoning is that she has wrongly assumed that a student's ability to learn mathematics is dependent on the time of day. Instead, it could be that Class A has smarter people and so naturally their average would be higher.

This is answer (c).

29.

Dr Wilson believes that the best way to lose weight is to perform fasted cardio where people do cardio as soon as they wake up in the morning and before they have eaten or drank anything. Dr Wilson conducted a 3-month study where his results indicated that individuals that did fasted cardio lost 3 kg more on average than individuals that did not do fasted cardio if they were eating healthy. Hence, he believes the best way to lose weight is to do fasted cardio alongside maintaining a healthy lifestyle.

1. Which of the following statements, if true, most strengthens Dr Wilson's argument?

☐

People who do cardio after eating in the morning tend to have a better mood throughout the day

☒

Fasted cardio burns fat directly because there is no glucose in the body from food leading to accelerated weight loss

✓

☐

Many of Dr Wilson's clients who do fasted cardio have found their weight stops changing after 4 months of using this method

☐

Dr Wilson's research study was applauded by many members of the scientific community

Question Feedback:

Dr Wilson states that fasted cardio is the best way to lose weight as soon as they wake up in the morning. Option (a) weakens our argument as it states eating before cardio can be more beneficial for mood. Option (c) weakens the argument. Option (d) slightly strengthens the argument by showing that others agree with his study. Option (b) strengthens the argument the most as it provides reasoning to why weight loss is greater in fasted cardio.

Hence, answer (b) helps in strengthening Dr Wilson's claim the most.

30.

Six friends are collecting stamps. They are putting them in albums that each hold 30 stamps. At the end of the collecting time, they want to ensure that they all have the same number of stamps. Five of them have 26 stamps per person - four short of filling their album - but the youngest one is lagging behind with her total. Each of the others gives her two stamps from their albums to equalise the numbers collected.

1. How many stamps did the youngest friend collect for herself?

1/1

1/1

☐ 6☐ 12☒ 14☐ 24**Question Feedback:**

Each album can hold 30 stamps.

Five friends are four short of filling their albums, which means they each have 26 stamps.

The youngest friend is behind, so each of the other five friends gives her some stamps.

Calculate the number of stamps the youngest friend has:

5×2 stamps (given by the other five friends) + her collection = 10 + her collection.

Since everyone's totals are equalised after giving away two cards, each friend has 24 cards. Therefore, the youngest friend collected 14 cards herself.

The solution therefore is (c).

31.

The lion is known as the 'King of the Jungle' because it is thought to be the strongest animal in the jungle. However, a tiger on average is larger and stronger than a lion. A lion can weigh up to 300 lbs while a tiger can weigh up to 500 lbs and there are more tigers in the African Jungle than there are lions. Clearly, the tiger is more worthy of being known as the 'King of the Jungle.'

1. Which of the following statements, if true, most strengthens the above argument?

☐ In research studies, lions have been shown to easily defeat tigers in a confrontation☐ There are more lions than tigers in every jungle other than the African jungle

1/1



In ecosystems where both lions and tigers exist, the tiger is often at the top of the food chain



The average African elephant is larger than the largest lions and tigers

Question Feedback:

The argument is that tigers are more deserving of being the "King of the Jungle". Option (a) weakens the argument. Option (b) also weakens the argument, while option (d) is irrelevant. Option (c) is the only one that strengthens the argument.

Hence, answer (c) best helps in strengthening the argument.

32.

Larry, Matthew, Ned, and Paul each pick a different number between 1 and 100. They each give a hint about the number, however, one of them decides to lie in their hint.

Larry: "My number is double Matthew's."

Matthew: "My number is 44."

Ned: "My number is 10 less than Matthew's."

Paul: "My number is 88."

1. Who could not have told the lie in their statement?

0/1



Larry



Matthew



Ned



Paul

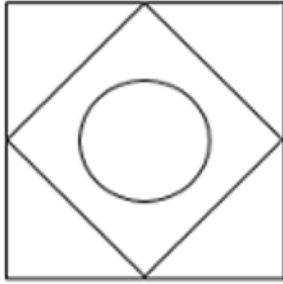
Question Feedback:

Since they all picked different numbers, there cannot be a repeat. If Larry, Ned, and Paul's statements are all true, then both Larry and Paul would have the same number (88). This means that one of them has to be a liar, leaving Ned to always be telling the truth.

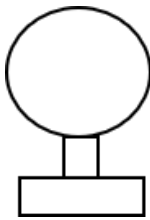
This is answer (c).

33.

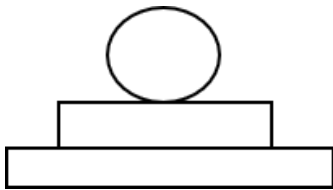
The top view of an object is shown below:



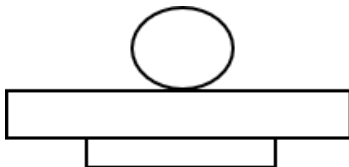
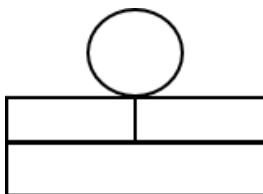
1. Which of the following could be the front view of this object?

☐

0/1

☒

✗

☐☐

✓

Question Feedback:

It cannot be answer (a) as the circle won't show the other two shapes from the top view if it is that big. It cannot be answer (c) either as the smaller rectangle cannot be seen from the top view. Answer (b) also doesn't work as the smaller rectangle needs to match the size of the bottom rectangle since the corners touch the edges of the bottom rectangle.

This shows that only answer (d) is correct.

34.

"A researcher needs to get money to run an experiment. If he is able to persuade sponsors during a presentation, he will immediately get the money he needs. Sponsors can only be persuaded by good presenters. The researcher is very insecure and believes he is bad at presentations, but he knows that everyone who attends 'Al's Presentation Workshop' will always become a good presenter by the end of the workshop."

Researcher: Therefore, if I go to Al's Presentation Workshop and then present to sponsors, I will be able to immediately get the money I need for my experiment.

1. If the quoted passage is true, which of the following shows the mistake the researcher has made?

0/1



The researcher may not actually be bad at presenting and only believes he is because of his insecurities



The researcher may need more money to do the experiment than the sponsors are willing to give him



The sponsors may require more than a good presentation to be persuaded to give their money to the researcher



The researcher may be the first person to fail 'Al's Presentation Workshop'

Question Feedback:

The mistake in the researcher's reasoning is that he has wrongly assumed the only way for the sponsors to accept him is with a good presentation. They may require more things to be convinced to sponsor the researcher.

This is answer (c).

35.

Five comedians (Patrice, Bill, Dave, Amy and Aries) perform their stand-up routines each on different weekdays.

- Patrice does not perform on Tuesday.
- Bill performs on Wednesday.
- Dave performs on a day after Patrice.
- Amy performs on Friday.

1. Who performs on Monday?

1/1

Patrice



Amy

Dave

Aries

Question Feedback:

	Monday	Tuesday	Wednesday	Thursday	Friday
Amy	No	No	No	No	Yes
Aries		Yes/No	No	Yes/No	No
Bill	No	No	Yes	No	No
Dave		Yes/No	No	Yes/No	No
Patrice	Yes	No	No		No

We know that Patrice is on Monday since Dave must perform on a day after Patrice. Since Amy has already taken Friday, this means that Patrice cannot go on Thursday, leaving Monday as his only option.

This is answer (a).

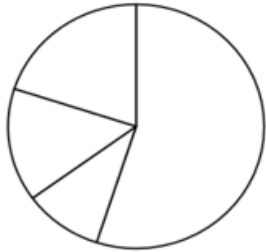
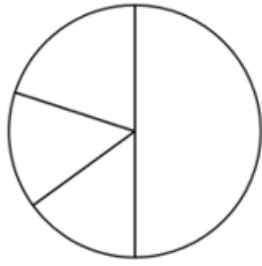
36.

Gary is a local farmer with 20 birds in total on his farm. These are either geese, ducks, eagles, or swans. The following is known about the birds:

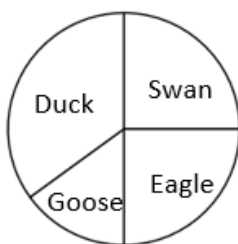
- The number of geese and ducks combined is five more than the number of eagles.
- The number of eagles and swans combined is the same as the number of geese and ducks combined.
- The number of eagles and ducks combined is seven more than the number of swans.

1. Which of these pie charts could represent the distribution of ducks, geese, eagles and swans on Gary's farm?

1/1



Question Feedback:

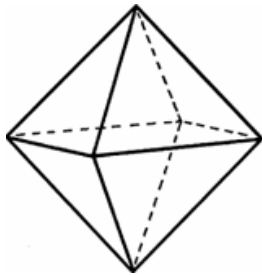


When we consider the information, this is the most correct pie chart as it satisfies the conditions given in the original question stem. We can figure it out using the second dot point, by looking for two sections when added together that take up half the pie chart. This is because they should be equal to the other two sections added together.

This is answer (d).

37.

An octahedron is a 3D shape made entirely of eight equilateral triangles. The diagram below shows an octahedron:



Sarah attempts to draw a net of the octahedron. She makes 3 attempts and the nets she drew for each attempt are shown below.

1/1

Attempt 1



Attempt 2



Attempt 3



1. How many of her nets can be used to successfully make an octahedron?

☐

0

☐

1

☒

2



☐

3

Question Feedback:

Attempts 1 and 2 when closed can form the octahedron. Attempt 3 however cannot close nicely and there is an overlap when trying to close the net of attempt 3.

This shows that 2 of them work which is answer (c).

38.

Even though, the use of fossil fuels has increased in the past 10 years, there have been no major natural disasters and the hole in the ozone layer has remained the same size. Therefore, fossil fuels have not damaged the ozone layer or harmed the environment.

1/1

1. Which of the following statements, if true, most weakens the above argument?

☐

Approximately 3 million tonnes of greenhouse gases have been produced over the last 10 years

☐

There is no evidence proving that fossil fuels impact the ozone layer

☒

The hole in the ozone layer would have shrunk had fossil fuels not been used over the past 10 years

☐

Climate change is scientifically proven to be true

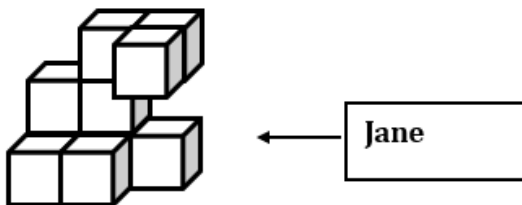
Question Feedback:

The argument is that fossil fuels have caused no damage to the environment in the last decade. Options (a) and (d) are irrelevant. Option (b) strengthens the argument. Option (c) is the only one that weakens the argument.

Hence, the answer is option (c) which helps in weakening the original argument.

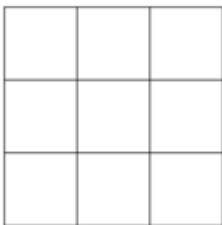
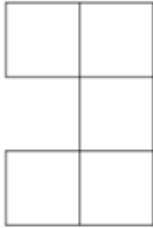
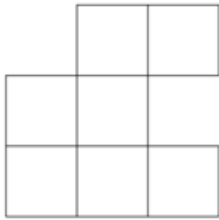
39.

Monty has created a stack of cubes which he shows to Jane. Jane views the stack from a particular side. A diagram of this is shown below:



1. What does the stack of cubes look like from Jane's perspective?

1/1

**Question Feedback:**

From Jane's perspective she would see this as the other blocks are hidden from her view:

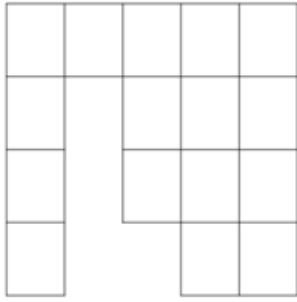


This is answer (b).

40.

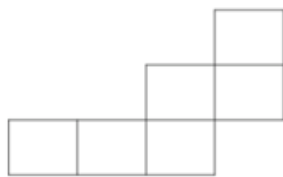
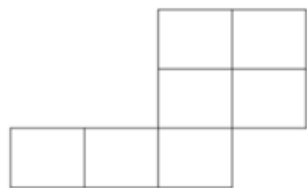
Caleb's father buys him a puzzle set for Christmas. The puzzle consists of three large pieces.

The final completed puzzle is shown in the diagram below:

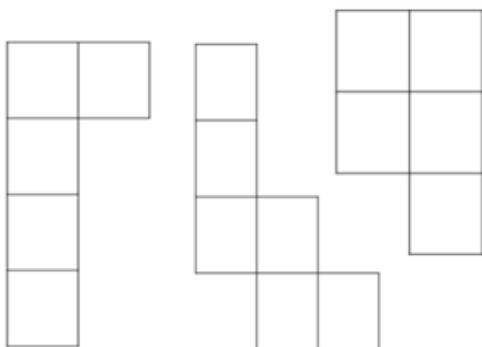


Three of the following pieces are part of the puzzle set.

1. Which of the following is NOT part of the puzzle set?

☐☐☐☒

Question Feedback:



This shows the 3 pieces before they were assembled.

This means that answer (d) is not a part of the puzzle.

CLOSE